COMPUTER APPLICATIONS

Maximum Marks: 100

Time allowed : Two hours

1. *Answers to this Paper must be written on the paper provided separately.*
2. *You will **not** be allowed to write during the first **15** minutes.*
3. *This time is to be spent in reading the question paper.*
4. ***The time given at the head of this Paper is the time allowed for writing the answers.***

5. *This Paper is divided into **two** Sections.*
6. *Attempt **all** questions from **Section A** and **any four** questions from **Section B.***
7. *The intended marks for questions or parts of questions are given in brackets[ ].*

Instruction for the Supervising Examiner

Kindly read aloud the Instructions given above to all the candidates present in the Examination Hall.

This paper consists of 12 printed pages.

SECTION A (40 Marks)

*(Attempt **all** questions from this **Section**.)*

Question 1 [20]

Choose the correct answers to the questions from the given options.

(Do not copy the questions, write the correct answers only.)

(i) What is the ***result*** of the expression: 7 * 8 % 3?

(a) 18

(b) 2

(c) 0

(d) 28

(ii) Which of the following ***Character methods results in char***?

(a) Character.isUpperCase(char)

(b) Character.isLowerCase(char)

(c) Character.toUpperCase(char)

(d) character.toupperCase(char

(iii) “4578” → 4578

Select the correct method to perform the above conversion:

(a) Integer.ParseInt(4578)

(b) Integer.parseInt(“4578”)

(c) Integer.parse(4578)

(d) Integer.parseint(“4578”)

(iv) Which of the following ***String methods*** can be ***Overloaded***?

(a) compareTo

(b) equalsIgnoreCase

(c) length

(d) substring

(v) The ***number of bytes*** occupied by the array of 10 elements which are decimal values.

(a) 80 bytes

(b) 320 bytes

(c) 10 bytes

(d) 8 bytes

(vi) The ***right statement to invoke the method*** perform which accepts an integer and a character and returns a double value.

(a) perform (45, ‘A’)

(b) double m = perform(45, ‘A’)

(c) double m = perform(“A”, 45)

(d) perform (‘A’, 45)

(vii) A method which has the ***same name of the class and initialises the instance variables with their default values is***:

(a) Parameterised Constructor

(b) Overloaded method

(c) Default constructor

(d) Non-parameterised constructor

(viii) Consider the following program segment:

```
if ( x > y )
System.out.println("Precious");
else
System.out.println("Priceless");
```

The equivalent statement for the above program segment with ***ternary operators*** is:

(a) x>y?System.out.println("Priceless"):System.out.println("Precious");

(b) System.out.println(x>y?"Priceless": "Precious");

(c) x>y?System.out.println("Precious"):System.out.println("Priceless");

(d) System.out.println(x>y? "Precious":"Priceless");

(ix) Consider the array : int x [ ] = {4, 3, 7, 9 , 120, -12, 8};

What is the ***index*** of the ***negative number*** of the array?

(a) 5

(b) 6

(c) 4

(d) x.length

(x) The output of the statement **"MEDITERRIAN".indexOf('R') is**

(a) true

(b) 7

(c) -1

(d) 6

(xi) Which of the following represents the ***escape sequence*** for a ***new line***?

(a) \n

(b) /n

(c) //n

(d) \\n

(xii) Raj was asked to accept the ***name of the employee, gender and the basic salary***. Help him to choose the ***correct datatypes in order***.

(a) String, char, double

(b) char, int, boolean

(c) double, char, String

(d) int, char, double

(xiii) Which statement is executed ***only once*** in a ***for*** statement?

```
for(initialisation;check;update)
{ body of the loop
}
```

(a) Update

(b) check

(c) initialisation

(d) body of the loop

(xiv) Fill in the blank in the following java statement.

Scanner ob = new Scanner (System.in);

char ch = ________________________;

(a) ob.nextLine( )

(b) ob.next().charAt()

(c) ob.next().charAt(0)

(d) ob.next().charat(0)

(xv) The statement to ***create an object*** "BridalGown" of the class "Apparel" with an integer argument as 34 is:

(a) Apparel BridalGown = new Apparel(34);

(b) BridalGown Apparel= new Apparel(34);

(c) Apparel BridalGown = new Apparel( );

(d) Apparel BridalGown = new BridalGown(34);

(xvi) Which of the following is the correct ***output*** for the java statement:

System.out.println("Dear Child\n\"May God Bless You\"");

(a) Dear Child
"May God Bless You"

(b) Dear Child"May God Bless You"

(c) Dear childMayGod Bless You

(d) Dear Child
\"May God Bless You\"

(xvii) The following character array is sorted in alphabetical order using ***SELECTION SORT*** method.

W	S	L	A	H

After the first iteration the array is changed to

A	S	L	W	H

What will be the array after the ***second iteration***?

(a)

W	S	L	A	H

(b)

A	H	L	A	S

(c)

A	H	L	W	S

(d)

W	E	L	A	H

(xviii) **Assertion (A):** Math methods are a part of the package java.lang.

Reason (R): Math method should be prefixed with the class name.

(a) (A) is true and (R) is false.

(b) (A) is false and (R) is true.

(c) Both (A) and (R) are true.

(d) Both (A) and (R) are false.

(xix) Which of the following is ***NOT*** a keyword?

(a) if

(b) for

(c) return

(d) Switch

(xx) Consider the array : int x[ ] [ ] = {{2, 3 ,4 }, { 7, 9 , 10}, {-11, 4, 6}};

What is the result of the statement : **System.out.println(x[0][2]*x[2][0]);**

(a) 24

(b) -44

(c) 16

(d) -22

Question 2

(i) ***Evaluate*** the following java expression: [2]

$x-=x++\ +\ ++x-x+x;$

When $x = 7$

(ii) Sam was asked to calculate the value of **m raised to the power of $\frac{2}{3}$**. She wrote the following code and the code resulted in **1.0** irrespective of the value of m. [2]

State the ***type of the error***, ***correct the code*** to obtain the correct answer:

System.out.println (Math.pow(m, 2/3));

(iii) Write the java expression for the following: [2]

Assign the ***sum of square root of M and the square of N to an appropriate variable.***

(iv) Differentiate between ***searching*** and ***sorting*** in single dimensional arrays. [2]

(v) Consider the following two dimensional array and answer the questions given below: [2]

char x[ ] [ ] = {(‘A’,’C’, ‘E’}, {‘M’, ‘A’, ‘P’}, {‘A’, ‘C’, ‘T’}};

(a) How ***many bytes*** are occupied by the array?

(b) What is the **index of ‘P’**?

(vi) Give the ***output*** of the following program segment, also mention how ***many times*** the following loop is executed. [2]

```
for ( k = 10;k > = 1 ; k - = 3 );
System.out.println(k*5);
```

(vii) Convert the following program segment using the ***Multiple branching statement***: [2]

```
if(a>=1&&a<=3)
System.out.println("Moderate");
else
System.out.println("Tough");
```

(viii) What is the ***output*** of the following statements: [2]

(a) "COMPASSIONATE".compareTo("COMPANION");

(b) Math.ceil(14.2) + Math.floor(17.9);

(ix) Write the ***prototype*** of the method CALCULATE which accepts two integers and a String and displays a formatted output. [2]

(x) ***Rewrite*** the following program segment using the ***exit controlled statement*** : [2]

```
for(k=1, m = 5;k<=10;k++, m++)
System.out.println(k*m);
```

SECTION B (60 Marks)

*(Answer **any four** questions from this **Section**.)*

The answers in this section should consist of the programs in BlueJ environment or any program environment with Java as the base.

*Each program should be written using variable description / mnemonic codes so that the logic of the program is clearly depicted. Flowcharts and algorithms are not required. Each question carries **15** marks.*

Question 3 [15]

Performance criteria of an employee is calculated based on the inputs as accuracy (a), reliability(r), and adaptability(ad) which is recorded as 1 to 10, 10 being the excellent score. Define a class with the following specifications:

Class name	:	performance
Member variables	:	String n – name of the employee int a, r, ad double basic, bonus
Member methods	:	
void input( )	–	to accept the name, accuracy(a), reliability(r) and adaptability(ad), basic from the user using the methods of Scanner class
void calculate ( )	–	to calculate the average of the three inputs (accuracy(a), reliability(r) and adaptability(ad)) and allot the bonus as per the given criteria:

$$average = \frac{accuracy+reliability+adaptibility}{3}$$

If average is above 7 then the bonus given to the employee is 7.5% of basic salary otherwise it is 5% of the basic salary.

void print ( )	–	to print the name of the employee, bonus.
void main ( )	–	to create an object of the class and invoke the methods.

Question 4 [15]

Define a class to accept a number and check if it is a ***Special buzz number or not***. A number is a Special buzz number ***if it ends with 7 and the sum of the digits of the number is also divisible by 7.***

Example: 957

The number ends with 7 and the sum of the digits (9 + 5 + 7 = 21) also is divisible by 7.

Question 5 [15]

Define a class to accept the names of 10 students in an array, arrange the names in alphabetical order using the ***BUBBLE SORT*** method, print the original array and the sorted array.

Question 6 [15]

A secret agency encodes the given message (String) as follows:

The first letter of the message is converted to @ if it is a Vowel, otherwise it is converted to #

All the other letters are reversed.

Print the original String and the encoded string.

Example : S = "APPEARANCE6"

Enoded String : @6ECNARAEPP

Example : S = "PERFORM"

Encoded String : #MROFRE

Question 7 [15]

Define a class to overload the method display as follows:

void display ( ) –	to display the format using NESTED LOOPS only: 1 2 3 4 5 6 7 8 9
void display (int n, char ch):	to print the square root of n if ch is 'S' or print the cube root of n if ch is 'C'
double display (double r, double h):	to return the volume of a cylinder as follows volume = area x h area = πr^2

Question 8 [15]

Define a class to accept values into a 4 x 4 array and check if it is a ***Super array or not***, an array is Super ***if the sum of even elements is greater than the sum of odd elements of the array.***

12	7	8	13
1	3	6	5
15	2	9	10
4	11	14	16

Sum of even elements = 12 + 8 + 6 + 2 + 10 + 4 + 14 + 16 = 72

Sum of odd elements = 7 + 13 + 1 + 3 + 5 + 15 + 9 + 11 = 64

Array is a super array.